



MATH230: Partial Derivatives

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Learning Objectives

- Calculate the mean, median, mode, and range of a data set
- Compute sample variance and standard deviation
- Construct quartiles, the IQR, and identify outliers
- Describe the shape, center, and spread of a distribution

Simplify each expression completely. Show all steps and circle your final answer.

Partial derivatives

1. Let $f(x,y) = 1x^2y^3$. Find the partial derivative $f_x(2,1)$.

$$f(x,y) = 1x^2y^3, (x_0, y_0) = (2, 1)$$

Answer: _____

2. The temperature at point (x,y) on a metal plate is $T(x,y) = 1x^3y^1$ degrees. Find the rate of change in temperature with respect to x at $(2,2)$.

$$f(x,y) = 1x^3y^1, (x_0, y_0) = (2, 2)$$

Answer: _____

3. Let $f(x,y) = 4x^3y^2$. Find the partial derivative $f_y(1,2)$.

$$f(x,y) = 4x^3y^2, (x_0, y_0) = (1, 2)$$

Answer: _____

4. Production $P(K,L) = 3K^2L^3$. Find the marginal product of labor: $P_L(2,1)$.

$$f(x,y) = 3x^2y^3, (x_0, y_0) = (2, 1)$$

Answer: _____

5. Let $f(x,y) = 1x^3y^1$. Find the partial derivative $f_x(3,3)$.

$$f(x,y) = 1x^3y^1, (x_0, y_0) = (3, 3)$$

Answer: _____



6. The temperature at point (x,y) on a metal plate is $T(x,y) = 3x^2y^2$ degrees. Find the rate of change in temperature with respect to x at $(3,2)$.

$$f(x,y) = 3x^2y^2, (x_0, y_0) = (3, 2)$$

Answer: _____

7. Let $f(x,y) = 5x^2y^2$. Find the partial derivative $f_y(3,3)$.

$$f(x,y) = 5x^2y^2, (x_0, y_0) = (3, 3)$$

Answer: _____

8. Production $P(K,L) = 1K^2L^3$. Find the marginal product of labor: $P_L(1,2)$.

$$f(x,y) = 1x^2y^3, (x_0, y_0) = (1, 2)$$

Answer: _____

9. Let $f(x,y) = 4x^3y^2$. Find the partial derivative $f_x(3,1)$.

$$f(x,y) = 4x^3y^2, (x_0, y_0) = (3, 1)$$

Answer: _____

10. The temperature at point (x,y) on a metal plate is $T(x,y) = 2x^2y^1$ degrees. Find the rate of change in temperature with respect to x at $(1,3)$.

$$f(x,y) = 2x^2y^1, (x_0, y_0) = (1, 3)$$

Answer: _____

11. Let $f(x,y) = 3x^1y^3$. Find the partial derivative $f_y(2,2)$.

$$f(x,y) = 3x^1y^3, (x_0, y_0) = (2, 2)$$

Answer: _____

12. Production $P(K,L) = 2K^2L^3$. Find the marginal product of labor: $P_L(3,3)$.

$$f(x,y) = 2x^2y^3, (x_0, y_0) = (3, 3)$$

Answer: _____

13. Let $f(x,y) = 2x^3y^3$. Find the partial derivative $f_x(2,2)$.

$$f(x,y) = 2x^3y^3, (x_0, y_0) = (2, 2)$$

Answer: _____



14. The temperature at point (x,y) on a metal plate is $T(x,y) = 1x^2y^1$ degrees. Find the rate of change in temperature with respect to x at $(3,2)$.

$$f(x, y) = 1x^2y^1, (x_0, y_0) = (3, 2)$$

Answer: _____

15. Let $f(x,y) = 5x^1y^3$. Find the partial derivative $f_y(3,3)$.

$$f(x, y) = 5x^1y^3, (x_0, y_0) = (3, 3)$$

Answer: _____

16. Production $P(K,L) = 1K^2L^2$. Find the marginal product of labor: $P_L(1,3)$.

$$f(x, y) = 1x^2y^2, (x_0, y_0) = (1, 3)$$

Answer: _____

17. Let $f(x,y) = 5x^2y^1$. Find the partial derivative $f_x(3,3)$.

$$f(x, y) = 5x^2y^1, (x_0, y_0) = (3, 3)$$

Answer: _____

18. The temperature at point (x,y) on a metal plate is $T(x,y) = 2x^2y^2$ degrees. Find the rate of change in temperature with respect to x at $(1,1)$.

$$f(x, y) = 2x^2y^2, (x_0, y_0) = (1, 1)$$

Answer: _____

19. Let $f(x,y) = 3x^2y^2$. Find the partial derivative $f_y(2,1)$.

$$f(x, y) = 3x^2y^2, (x_0, y_0) = (2, 1)$$

Answer: _____

20. Production $P(K,L) = 4K^1L^2$. Find the marginal product of labor: $P_L(2,2)$.

$$f(x, y) = 4x^1y^2, (x_0, y_0) = (2, 2)$$

Answer: _____

21. Let $f(x,y) = 5x^2y^1$. Find the partial derivative $f_x(1,2)$.

$$f(x, y) = 5x^2y^1, (x_0, y_0) = (1, 2)$$

Answer: _____



22. The temperature at point (x,y) on a metal plate is $T(x,y) = 4x^3y^2$ degrees. Find the rate of change in temperature with respect to x at (2,2).

$$f(x, y) = 4x^3y^2, (x_0, y_0) = (2, 2)$$

Answer: _____

23. Let $f(x,y) = 4x^1y^3$. Find the partial derivative $f_y(3,3)$.

$$f(x, y) = 4x^1y^3, (x_0, y_0) = (3, 3)$$

Answer: _____

24. Production $P(K,L) = 4K^2L^3$. Find the marginal product of labor: $P_L(3,1)$.

$$f(x, y) = 4x^2y^3, (x_0, y_0) = (3, 1)$$

Answer: _____

25. Let $f(x,y) = 1x^3y^2$. Find the partial derivative $f_x(1,2)$.

$$f(x, y) = 1x^3y^2, (x_0, y_0) = (1, 2)$$

Answer: _____

26. The temperature at point (x,y) on a metal plate is $T(x,y) = 1x^2y^2$ degrees. Find the rate of change in temperature with respect to x at (2,2).

$$f(x, y) = 1x^2y^2, (x_0, y_0) = (2, 2)$$

Answer: _____

27. Let $f(x,y) = 4x^3y^2$. Find the partial derivative $f_y(3,3)$.

$$f(x, y) = 4x^3y^2, (x_0, y_0) = (3, 3)$$

Answer: _____

28. Production $P(K,L) = 1K^1L^2$. Find the marginal product of labor: $P_L(1,3)$.

$$f(x, y) = 1x^1y^2, (x_0, y_0) = (1, 3)$$

Answer: _____

29. Let $f(x,y) = 1x^3y^3$. Find the partial derivative $f_x(2,2)$.

$$f(x, y) = 1x^3y^3, (x_0, y_0) = (2, 2)$$

Answer: _____



30. The temperature at point (x,y) on a metal plate is $T(x,y) = 3x^2y^2$ degrees. Find the rate of change in temperature with respect to x at $(1,1)$.

$$f(x, y) = 3x^2y^2, (x_0, y_0) = (1, 1)$$

Answer: _____





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ANSWER KEY & SOLUTIONS

Topics: Partial derivatives. All answers verified by independent computation.

Solutions



Partial derivatives

1. Let $f(x,y) = 1x^2y^3$. Find the partial derivative $f_x(2,1)$.

$$f(x,y) = 1x^2y^3, (x_0, y_0) = (2, 1)$$

→ Differentiate with respect to x , treating y as constant.

$$\rightarrow f_x = 2x^1y^3.$$

$$\rightarrow f_x(2,1) = 2 \cdot 2^1 \cdot 1^3 = 4.$$

Answer: $f_x = 2x^1y^3, f_x(2, 1) = 4$

2. The temperature at point (x,y) on a metal plate is $T(x,y) = 1x^3y^1$ degrees. Find the rate of change in temperature with respect to x at $(2,2)$.

$$f(x,y) = 1x^3y^1, (x_0, y_0) = (2, 2)$$

→ $T_x = 3x^2y^1$ (rate of change in x -direction).

$$\rightarrow T_x(2,2) = 24 \text{ degrees per unit.}$$

Answer: $f_x = 3x^2y^1, f_x(2, 2) = 24$

3. Let $f(x,y) = 4x^3y^2$. Find the partial derivative $f_y(1,2)$.

$$f(x,y) = 4x^3y^2, (x_0, y_0) = (1, 2)$$

→ Differentiate with respect to y , treating x as constant.

$$\rightarrow f_y = 8x^3y^1.$$

$$\rightarrow f_y(1,2) = 8 \cdot 1^3 \cdot 2^1 = 16.$$

Answer: $f_y = 8x^3y^1, f_y(1, 2) = 16$

4. Production $P(K,L) = 3K^2L^3$. Find the marginal product of labor: $P_L(2,1)$.

$$f(x,y) = 3x^2y^3, (x_0, y_0) = (2, 1)$$

→ Differentiate with respect to L (treating K as constant).

$$\rightarrow P_L = 9K^2L^2.$$

$$\rightarrow P_L(2,1) = 36 \text{ units per unit labor.}$$

Answer: $f_y = 9x^2y^2, f_y(2, 1) = 36$

5. Let $f(x,y) = 1x^3y^1$. Find the partial derivative $f_x(3,3)$.

$$f(x,y) = 1x^3y^1, (x_0, y_0) = (3, 3)$$

→ Differentiate with respect to x , treating y as constant.

$$\rightarrow f_x = 3x^2y^1.$$

$$\rightarrow f_x(3,3) = 3 \cdot 3^2 \cdot 3^1 = 81.$$

Answer: $f_x = 3x^2y^1, f_x(3, 3) = 81$



6. The temperature at point (x,y) on a metal plate is $T(x,y) = 3x^2y^2$ degrees. Find the rate of change in temperature with respect to x at $(3,2)$.

$$f(x,y) = 3x^2y^2, (x_0, y_0) = (3, 2)$$

→ $T_x = 6x^1y^2$ (rate of change in x -direction).

→ $T_x(3,2) = 72$ degrees per unit.

Answer: $f_x = 6x^1y^2, f_x(3, 2) = 72$

7. Let $f(x,y) = 5x^2y^2$. Find the partial derivative $f_y(3,3)$.

$$f(x,y) = 5x^2y^2, (x_0, y_0) = (3, 3)$$

→ Differentiate with respect to y , treating x as constant.

→ $f_y = 10x^2y^1$.

→ $f_y(3,3) = 10 \cdot 3^2 \cdot 3^1 = 270$.

Answer: $f_y = 10x^2y^1, f_y(3, 3) = 270$

8. Production $P(K,L) = 1K^2L^3$. Find the marginal product of labor: $P_L(1,2)$.

$$f(x,y) = 1x^2y^3, (x_0, y_0) = (1, 2)$$

→ Differentiate with respect to L (treating K as constant).

→ $P_L = 3K^2L^2$.

→ $P_L(1,2) = 12$ units per unit labor.

Answer: $f_y = 3x^2y^2, f_y(1, 2) = 12$

9. Let $f(x,y) = 4x^3y^2$. Find the partial derivative $f_x(3,1)$.

$$f(x,y) = 4x^3y^2, (x_0, y_0) = (3, 1)$$

→ Differentiate with respect to x , treating y as constant.

→ $f_x = 12x^2y^2$.

→ $f_x(3,1) = 12 \cdot 3^2 \cdot 1^2 = 108$.

Answer: $f_x = 12x^2y^2, f_x(3, 1) = 108$

10. The temperature at point (x,y) on a metal plate is $T(x,y) = 2x^2y^1$ degrees. Find the rate of change in temperature with respect to x at $(1,3)$.

$$f(x,y) = 2x^2y^1, (x_0, y_0) = (1, 3)$$

→ $T_x = 4x^1y^1$ (rate of change in x -direction).

→ $T_x(1,3) = 12$ degrees per unit.

Answer: $f_x = 4x^1y^1, f_x(1, 3) = 12$



11. Let $f(x,y) = 3x^1y^3$. Find the partial derivative $f_y(2,2)$.

$$f(x,y) = 3x^1y^3, (x_0, y_0) = (2, 2)$$

→ Differentiate with respect to y , treating x as constant.

$$\rightarrow f_y = 9x^1y^2.$$

$$\rightarrow f_y(2,2) = 9 \cdot 2^1 \cdot 2^2 = 72.$$

Answer: $f_y = 9x^1y^2, f_y(2, 2) = 72$

12. Production $P(K,L) = 2K^2L^3$. Find the marginal product of labor: $P_L(3,3)$.

$$f(x,y) = 2x^2y^3, (x_0, y_0) = (3, 3)$$

→ Differentiate with respect to L (treating K as constant).

$$\rightarrow P_L = 6K^2L^2.$$

$$\rightarrow P_L(3,3) = 486 \text{ units per unit labor.}$$

Answer: $f_y = 6x^2y^2, f_y(3, 3) = 486$

13. Let $f(x,y) = 2x^3y^3$. Find the partial derivative $f_x(2,2)$.

$$f(x,y) = 2x^3y^3, (x_0, y_0) = (2, 2)$$

→ Differentiate with respect to x , treating y as constant.

$$\rightarrow f_x = 6x^2y^3.$$

$$\rightarrow f_x(2,2) = 6 \cdot 2^2 \cdot 2^3 = 192.$$

Answer: $f_x = 6x^2y^3, f_x(2, 2) = 192$

14. The temperature at point (x,y) on a metal plate is $T(x,y) = 1x^2y^1$ degrees. Find the rate of change in temperature with respect to x at $(3,2)$.

$$f(x,y) = 1x^2y^1, (x_0, y_0) = (3, 2)$$

→ $T_x = 2x^1y^1$ (rate of change in x -direction).

$$\rightarrow T_x(3,2) = 12 \text{ degrees per unit.}$$

Answer: $f_x = 2x^1y^1, f_x(3, 2) = 12$

15. Let $f(x,y) = 5x^1y^3$. Find the partial derivative $f_y(3,3)$.

$$f(x,y) = 5x^1y^3, (x_0, y_0) = (3, 3)$$

→ Differentiate with respect to y , treating x as constant.

$$\rightarrow f_y = 15x^1y^2.$$

$$\rightarrow f_y(3,3) = 15 \cdot 3^1 \cdot 3^2 = 405.$$

Answer: $f_y = 15x^1y^2, f_y(3, 3) = 405$



16. Production $P(K,L) = 1K^2L^2$. Find the marginal product of labor: $P_L(1,3)$.

$$f(x,y) = 1x^2y^2, (x_0, y_0) = (1, 3)$$

→ Differentiate with respect to L (treating K as constant).

$$\rightarrow P_L = 2K^2L^1.$$

$$\rightarrow P_L(1,3) = 6 \text{ units per unit labor.}$$

Answer: $f_y = 2x^2y^1, f_y(1, 3) = 6$

17. Let $f(x,y) = 5x^2y^1$. Find the partial derivative $f_x(3,3)$.

$$f(x,y) = 5x^2y^1, (x_0, y_0) = (3, 3)$$

→ Differentiate with respect to x , treating y as constant.

$$\rightarrow f_x = 10x^1y^1.$$

$$\rightarrow f_x(3,3) = 10 \cdot 3^1 \cdot 3^1 = 90.$$

Answer: $f_x = 10x^1y^1, f_x(3, 3) = 90$

18. The temperature at point (x,y) on a metal plate is $T(x,y) = 2x^2y^2$ degrees. Find the rate of change in temperature with respect to x at $(1,1)$.

$$f(x,y) = 2x^2y^2, (x_0, y_0) = (1, 1)$$

→ $T_x = 4x^1y^2$ (rate of change in x -direction).

$$\rightarrow T_x(1,1) = 4 \text{ degrees per unit.}$$

Answer: $f_x = 4x^1y^2, f_x(1, 1) = 4$

19. Let $f(x,y) = 3x^2y^2$. Find the partial derivative $f_y(2,1)$.

$$f(x,y) = 3x^2y^2, (x_0, y_0) = (2, 1)$$

→ Differentiate with respect to y , treating x as constant.

$$\rightarrow f_y = 6x^2y^1.$$

$$\rightarrow f_y(2,1) = 6 \cdot 2^2 \cdot 1^1 = 24.$$

Answer: $f_y = 6x^2y^1, f_y(2, 1) = 24$

20. Production $P(K,L) = 4K^1L^2$. Find the marginal product of labor: $P_L(2,2)$.

$$f(x,y) = 4x^1y^2, (x_0, y_0) = (2, 2)$$

→ Differentiate with respect to L (treating K as constant).

$$\rightarrow P_L = 8K^1L^1.$$

$$\rightarrow P_L(2,2) = 32 \text{ units per unit labor.}$$

Answer: $f_y = 8x^1y^1, f_y(2, 2) = 32$



21. Let $f(x,y) = 5x^2y^1$. Find the partial derivative $f_x(1,2)$.

$$f(x,y) = 5x^2y^1, (x_0, y_0) = (1, 2)$$

→ Differentiate with respect to x , treating y as constant.

$$\rightarrow f_x = 10x^1y^1.$$

$$\rightarrow f_x(1,2) = 10 \cdot 1^1 \cdot 2^1 = 20.$$

Answer: $f_x = 10x^1y^1, f_x(1, 2) = 20$

22. The temperature at point (x,y) on a metal plate is $T(x,y) = 4x^3y^2$ degrees. Find the rate of change in temperature with respect to x at $(2,2)$.

$$f(x,y) = 4x^3y^2, (x_0, y_0) = (2, 2)$$

→ $T_x = 12x^2y^2$ (rate of change in x -direction).

$$\rightarrow T_x(2,2) = 192 \text{ degrees per unit.}$$

Answer: $f_x = 12x^2y^2, f_x(2, 2) = 192$

23. Let $f(x,y) = 4x^1y^3$. Find the partial derivative $f_y(3,3)$.

$$f(x,y) = 4x^1y^3, (x_0, y_0) = (3, 3)$$

→ Differentiate with respect to y , treating x as constant.

$$\rightarrow f_y = 12x^1y^2.$$

$$\rightarrow f_y(3,3) = 12 \cdot 3^1 \cdot 3^2 = 324.$$

Answer: $f_y = 12x^1y^2, f_y(3, 3) = 324$

24. Production $P(K,L) = 4K^2L^3$. Find the marginal product of labor: $P_L(3,1)$.

$$f(x,y) = 4x^2y^3, (x_0, y_0) = (3, 1)$$

→ Differentiate with respect to L (treating K as constant).

$$\rightarrow P_L = 12K^2L^2.$$

$$\rightarrow P_L(3,1) = 108 \text{ units per unit labor.}$$

Answer: $f_y = 12x^2y^2, f_y(3, 1) = 108$

25. Let $f(x,y) = 1x^3y^2$. Find the partial derivative $f_x(1,2)$.

$$f(x,y) = 1x^3y^2, (x_0, y_0) = (1, 2)$$

→ Differentiate with respect to x , treating y as constant.

$$\rightarrow f_x = 3x^2y^2.$$

$$\rightarrow f_x(1,2) = 3 \cdot 1^2 \cdot 2^2 = 12.$$

Answer: $f_x = 3x^2y^2, f_x(1, 2) = 12$



26. The temperature at point (x,y) on a metal plate is $T(x,y) = 1x^2y^2$ degrees. Find the rate of change in temperature with respect to x at $(2,2)$.

$$f(x,y) = 1x^2y^2, (x_0, y_0) = (2, 2)$$

→ $T_x = 2x^1y^2$ (rate of change in x -direction).

→ $T_x(2,2) = 16$ degrees per unit.

Answer: $f_x = 2x^1y^2, f_x(2, 2) = 16$

27. Let $f(x,y) = 4x^3y^2$. Find the partial derivative $f_y(3,3)$.

$$f(x,y) = 4x^3y^2, (x_0, y_0) = (3, 3)$$

→ Differentiate with respect to y , treating x as constant.

→ $f_y = 8x^3y^1$.

→ $f_y(3,3) = 8 \cdot 3^3 \cdot 3^1 = 648$.

Answer: $f_y = 8x^3y^1, f_y(3, 3) = 648$

28. Production $P(K,L) = 1K^1L^2$. Find the marginal product of labor: $P_L(1,3)$.

$$f(x,y) = 1x^1y^2, (x_0, y_0) = (1, 3)$$

→ Differentiate with respect to L (treating K as constant).

→ $P_L = 2K^1L^1$.

→ $P_L(1,3) = 6$ units per unit labor.

Answer: $f_y = 2x^1y^1, f_y(1, 3) = 6$

29. Let $f(x,y) = 1x^3y^3$. Find the partial derivative $f_x(2,2)$.

$$f(x,y) = 1x^3y^3, (x_0, y_0) = (2, 2)$$

→ Differentiate with respect to x , treating y as constant.

→ $f_x = 3x^2y^3$.

→ $f_x(2,2) = 3 \cdot 2^2 \cdot 2^3 = 96$.

Answer: $f_x = 3x^2y^3, f_x(2, 2) = 96$

30. The temperature at point (x,y) on a metal plate is $T(x,y) = 3x^2y^2$ degrees. Find the rate of change in temperature with respect to x at $(1,1)$.

$$f(x,y) = 3x^2y^2, (x_0, y_0) = (1, 1)$$

→ $T_x = 6x^1y^2$ (rate of change in x -direction).

→ $T_x(1,1) = 6$ degrees per unit.

Answer: $f_x = 6x^1y^2, f_x(1, 1) = 6$

